SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 –Industry Problem Solving	Assessment:	Assessment Tool 2– Industry Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	M.SREEVARDHAN	Reg. No.:	192472405
Section / Batch:	CSE-AI/2024	Date:	20-08-2026

Code of Conduct
I M.SREEVARDHAN/192472405 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: ___VAYU_______________________

Criteria	Weigh t age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Mark s
Understandi ng of Case	25	Thorough understandin g; clearly identifies all key issues	Good understandin g with most issues identified	Basic understandin g; some issues missed	Poor understandin g; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong	Good analysis with logical	Basic analysis with limited depth	Weak or no analysis; lacks	

		reasoning and insights	reasoning		reasoning	
Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

Problem 16: Cybersecurity - Detection Algorithm Improvement:

1. Population and Sample: Population: All potential malicious activities facing the system. Sample: 200 recent malicious attacks.

2. Parameter of Interest: The true proportion (p) of malicious activities correctly detected by the new algorithm.

3. Point Estimate: Sample proportion ($\hat{p}$) = 194 / 200 = 0.970 (97%).

4. Null Hypothesis (H): $p \le 0.95$ (The new algorithm is not better than the historical baseline).

5. Alternative Hypothesis (H): $p > 0.95$ (The new algorithm significantly improves detection).

6. Significance Level (α): 0.05 (Standard default).

7. Test Statistic: Z-statistic = 1.298

8. p-value: 0.0972

9. Compare p-value with α: 0.0971 > 0.05. We fail to reject the null hypothesis.

10. Industry-Oriented Conclusion (Delta Explanation): Under the High-Efficiency Protocol, the 'Black Box' output shows a slight positive delta (97% vs 95%), but Vector Embeddings of the statistical distribution indicate this variation is likely noise. We do not have sufficient evidence to confidently say the new algorithm is fundamentally superior. The cybersecurity team should collect more data or retrain the model before full deployment.

Problem 17: Healthcare - Appointment Scheduling Wait Times:

1. Population and Sample: Population: All patients using the new scheduling system. Sample: 64 patients who used the system.

2. Parameter of Interest: The true mean waiting time (μ) of patients under the new system.

3. Point Estimate: Sample mean ($\bar{x}$) = 38 minutes.

4. Null Hypothesis (H): $\mu \geq 42$ minutes (The new system does not reduce waiting time).

5. Alternative Hypothesis (H): $\mu < 42$ minutes (The new system significantly reduces waiting time).

6. Significance Level (α): 0.05 (Standard default).

7. Test Statistic: t-statistic = -2.667

8. p-value: 0.0049

9. Compare p-value with α: $0.0049 < 0.05$. We reject the null hypothesis.

10. Industry-Oriented Conclusion (Delta Explanation): The Recursive Feedback from the sample indicates a highly significant reduction in patient wait time. Applying Zero-Shot Inference on the core metrics confirms the operational bottleneck has been mitigated. The hospital administration is justified in rolling out this scheduling system permanently to enhance patient satisfaction.

Problem 18: Social Media Analytics - Daily Usage Time:

1. Population and Sample: Population: All active users of the social media platform. Sample: 100 randomly selected users.

2. Parameter of Interest: The true mean daily usage time (μ) per user.

3. Point Estimate: Sample mean ($\bar{x}$) = 3.2 hours.

4. Null Hypothesis (H): μ = 3 hours (The company's claim is accurate).

5. Alternative Hypothesis (H): μ ≠ 3 hours (The average usage time differs from 3 hours).

6. Significance Level (α): 0.05 (As specified).

7. Test Statistic: t-statistic = 2.500

8. p-value: 0.0141

9. Compare p-value with α: 0.0140 < 0.05. We reject the null hypothesis.

10. Industry-Oriented Conclusion (Delta Explanation): Chain-of-Thought Verification exposes that user engagement fundamentally differs from the assumed 3-hour baseline. The 80/20 Core Insight highlights an upward engagement delta. The platform should update its marketing and server-load estimations to reflect this higher, statistically verified 3.2-hour average usage.

Problem 19: Machine Learning - Classification Model Accuracy:

1. Population and Sample: Population: All possible predictions made by the new model. Sample: 500 randomly selected test observations.

2. Parameter of Interest: The true accuracy rate (p) of the new classification model.

3. Point Estimate: Sample proportion ($\hat{p}$) = 0.86 (86%).

4. Null Hypothesis (H): p ≤ 0.82 (The new model is equal to or worse than the existing 82% model).

5. Alternative Hypothesis (H): p > 0.82 (The new model has significantly improved accuracy).

6. Significance Level (α): 0.05 (Standard default).

7. Test Statistic: Z-statistic = 2.328

8. p-value: 0.0100

9. Compare p-value with α: 0.0099 < 0.05. We reject the null hypothesis.

10. Industry-Oriented Conclusion (Delta Explanation): The Attention Head Focusing on predictive outcomes proves the new algorithm represents a substantial architectural upgrade. Because the p-value is extremely low, the team has statistically verified the positive delta. The new classification model should be pushed to production immediately to replace the legacy system.

Problem 20: Manufacturing / Quality Control - Defect Rate:

1. Population and Sample: Population: All products produced by the current manufacturing process. Sample: 2,000 randomly selected products.

2. Parameter of Interest: The true defect rate proportion (p) of the production process.

3. Point Estimate: Sample proportion ($\hat{p}$) = 55 / 2000 = 0.0275 (2.75%).

4. Null Hypothesis (H): p = 0.02 (The defect rate is exactly 2% as claimed).

5. Alternative Hypothesis (H): $p \neq 0.02$ (The true defect rate differs from 2%).

6. Significance Level (α): 0.05 (As specified).

7. Confidence Interval and Test Statistic: 95% CI = (0.0203, 0.0347). Z-statistic = 2.396

8. p-value: 0.0166

9. Compare p-value with α: 0.0166 < 0.05. We reject the null hypothesis. Additionally, the 95% CI (2.03% to 3.47%) does not contain the claimed 2%.

10. Industry-Oriented Conclusion (Delta Explanation): Using Dual Chain-of-Thought Verification (CI + p-value), the Black Box output indisputably rejects the manufacturer's claim. Both methods are congruent: the true defect rate is significantly higher than the acceptable 2% threshold. Immediate quality control intervention and root-cause analysis are mandated to correct the manufacturing drift.